Multimodal Transportation Network Optimization with Environmental and Economic Performance Considered: An Ongoing Research

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***Abstract*—An optimization model is proposed in this paper to solve the transportation mode selection and the demand allocation problem with environmental and economic performance considered. The case study based on the regional logistics of Pearl River Delta (PRD) in Guangdong province of China illustrates that the optimal solution changes significantly when carbon emission is introduced into the optimization objective. More ocean and less ordinary road transportation will be adopted when environmental performance matters, and it is possible to greatly reduce the carbon emission while keeping the transportation cost not increase dramatically. However, there exist a tradeoff between emission reduction and the transportation cost. The proposed model can also be extended to take the transportation time and due dates of demands into consideration in the process of multimodal transportation network optimization.**

Keywords—Multimodal transportation; Carbon emissions; Network optimization

I. INTRODUCTION

With the development of economy, the transportation of production material and product is more and more frequent, which results in that the transportation cost becomes one of the main parts of enterprise cost (Song et al. 2014). Therefore, Multimodal transportation that combines different transportation modes such as ordinary road, highway, railway, waterway, ocean shipment and airway is regarded as an effective and efficient solution to reduce transportation cost (Kumar et al. 2013). However, carbon emission coming from the energy consumption of transportation also plays an important role in the greenhouse gas emission, which is considered by many scientists as a key factor leading to the current environmental problems such as global warming (Wang and He 2017). As a result, more and more research on transportation including logistics and supply chain has taken both environmental performance (e.g. carbon emission) and economic performance (e.g. cost) into consideration in their models (Du et al. 2015).

After the investigation, Bouchery et al. (2014) indicated that multimodal transportation may also be a way to reduce carbon emission without "damaging" the economic growth. Guo (2007) takes the transportation cost and air pollution cost as the objective function for optimization. Under the condition of determined demands, a simulated annealing algorithm was developed to solve both the problem of transportation selection among ordinary road, highway and railway and the problem of vehicle routing. Based on low-carbon economy, Huang and Lin (2011) compared the carbon emissions between road and rail transportation and proposed a possible low-carbon transportation way for container, i.e. integrating railway with ocean shipment to form the multimodal transportation. Regarding the airport as a node of multimodal transportation, Janic (2011) evaluated the impact of transportation network linking airport and highway on environmental performance. Makhloufi et al. (2015) developed a new discrete event simulator called MODUMS to evaluate the economic and environmental impact of a logistic system. Jiang et al. (2012) measured the carbon emissions and energy consumption of three different multimodal transportation modes in harbor by using the IPCC methodology. They also did a case study in their paper based on Shanghai harbor. Chen et al. (2015) built a multimodal transportation optimization model under different emission reduction regulations such as carbon tax and cap-and-trade. Their research shows that both the allocated emission allowance and the price of emission permit have significant impact on the selection of transportation modes. However, cap-and-trade scheme provides more flexibility for the operations of generating companies such as logistics enterprises.

Although environmental performance is considered as well as economic performance in the above papers, few of them integrate transportation mode selection with routing optimization and demand allocation. As a matter of fact, due to the change of transportation mode and route, the optimal demand allocation policy, i.e. supply and demand combination also changes. In this paper, we try to propose a general model for decision making of demand allocation integrating multimodal transportation with environmental and economic performance considered. A more real-life case study based on the regional logistics of Pearl River Delta (PRD) in Guangdong province of China is also studied. Though it is still an ongoing research, some exciting results have been obtained to show the significant impact on demand allocation and transportation mode selection when environmental performance is taken into consideration.

The remainder of this paper is organized as follows. Section II describes the problem and proposes the general optimization model. Section III demonstrates the case study and the

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experiment results. The conclusion and discussion are then provided in the last section.

II. PROBLEM DESCRIPTION AND MODELLING

Given a transportation network G(N, V), N is the node set and V the arc set. In the node set, there are three subsets, i.e. demand node set N_q, supply node set N_s and transshipment node set N_t. An arc is denoted by three factors (i, j, m), where i is the beginning node and j the ending node, and m is the transportation mode. Different transportation mode leads to different unit transportation cost and carbon emission. Suppose the quantity of demands and supplies are already known and the total supply is enough for satisfying the total demand. In a regional transportation network, the supply nodes and the demand nodes may also be used to deliver goods like transshipment nodes. The problem is to determine how to satisfy the demands, namely, deliver how much goods via each arc including beginning node, ending node and transportation mode, so that the total transportation cost and the total carbon emission are minimized.

To model the problem, the following notations are defined.

$i, j \in N$. The index of network node.

$k \in N_s$. The index of supply node.

$l \in N_q$. The index of demand node.

$n \in N_t$. The index of transshipment node.

m. The index of transportation mode.

q_l. The demand quantity of demand node l.

s_k. The supply quantity of supply node k.

d_{ijm}. The distance from node i to node j via transportation mode m.

c_{ijm}. The unit transportation cost from node i to node j via transportation mode m.

e_{ijm}. The unit carbon emission from node i to node j via transportation mode m.

y_{ijm}. The transportation quantity (decision variable) from node i to node j via transportation mode m. If $y_{ijm} > 0$, the transportation mode m from node i to node j is selected; otherwise, if $y_{ijm} = 0$, then the transportation mode m from node i to node j is not selected.

The objective functions of the problem can be constructed as follows.

$$\min f_1 = \sum_{(i,j,m)\in A} c_{ijm} \cdot y_{ijm} \quad (1)$$

$$\min f_2 = \sum_{(i,j,m)\in A} e_{ijm} \cdot y_{ijm} \quad (2)$$

Subject to the following constraints.

$$\sum_{(i,l,m)\in A} y_{ilm} = q_l + \sum_{(l,j,m)\in A} y_{ljm}, \forall l \in N_q \quad (3)$$

$$\sum_{(k,j,m)\in A} y_{kjm} \le s_k + \sum_{(i,k,m)\in A} y_{ikm}, \forall k \in N_s \quad (4)$$

$$\sum_{(i,n,m)\in A} y_{inm} = \sum_{(n,j,m)\in A} y_{njm}, \forall n \in N_t \quad (5)$$

$$y_{ijm} \ge 0, \forall (i, j, m) \in A \quad (6)$$

The first objective is to minimize the economic performance of total transportation cost, while the second objective is environmental performance of total carbon emission. Constraint (3) assures that for any demand node, the input is equal to the demand plus the transshipment quantity. Due to the assumption that total supply is larger than or equal to the total demand, constraint (4) illustrates the supply quantity of a supply node plus its transshipment/input quantity should be not less than its total output quantity. Constraint (5) keeps the balance of a transshipment node. The last constraint makes sure the transportation quantity in each arc is nonnegative.

The above presents a general optimization model for transportation mode selection and demand allocation with economic and environmental performance considered. For extension, the two objective functions can also be combined into one by averaging or weighted averaging. Other objectives or constraints such as the due dates of demands and penalty cost when transportation delay appears can be added into the model too.

III. A CASE STUDY

In this section, a case based on the regional logistics of Pearl River Delta (PRD) in Guangdong province of China is studied. In the case study, enterprise ZC existed in the paper-making industry produces products from its four factories located in Guangzhou, Foshan, Dongguan and Shanwei, and delivers the products to five customer zones including Shenzhen, Zhuhai, Huizhou, Zhongshan and Jiangmen. The weekly quantities of supplies from Guangzhou, Foshan, Dongguan and Shanwei are 800, 1200, 311, 800 tons respectively, and the demands are 920, 576, 461, 354, and 800 tons respectively in Shenzhen, Zhuhai, Huizhou, Zhongshan and Jiangmen. Figure 1 illustrates the transportation network, in which twenty one (21) nodes, four transportation modes (i.e. ordinary road, highway, railway and ocean shipment) and more than fifty (50) arcs are included. The unit transportation cost and the unit carbon emission of different transportation mode is shown in Table 1. The transportation cost of ordinary road is estimated based on the consumption of diesel and its price; while the transportation cost of highway has to take the toll fee into consideration besides the cost of diesel. According to the data of China railway (see website kyfw.12306.cn/otn/), besides the unit transportation cost in terms of mileage and weight (0.155 Chinese yuan/ton·km), a base cost of 28 yuan/ton should also be added into the railway transportation cost. Based on the analysis of Chen et al. (2010), the unit transportation cost by ocean is estimated at 0.25 yuan/ton·km. The estimation of unit carbon emission is based on both energy consumption and emission coefficient of the transportation mode.

According to the above mathematical model and data, IBM ILOG CPLEX© is used to implement the computing for the case study. We first study the single objective problem, and then extend it to multiple objective situation by ε-constraint method (Ehrgott and Ruzika, 2008), i.e. transform the objective of minimizing total emission into constraint.

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Table 1. THE UNIT TRANSPORTATION COST AND UNIT CARBON EMISSION OF DIFFERENT TRANSPORTATION MODE

	Ordinary road	Highway	Railway	Ocean shipment
Unit transportation cost (yuan/ton·km)	2.178	2.365	0.155	0.250
Unit carbon emission (10^{-5} ton/ton·km)	87.770	78.990	1.468	1.897

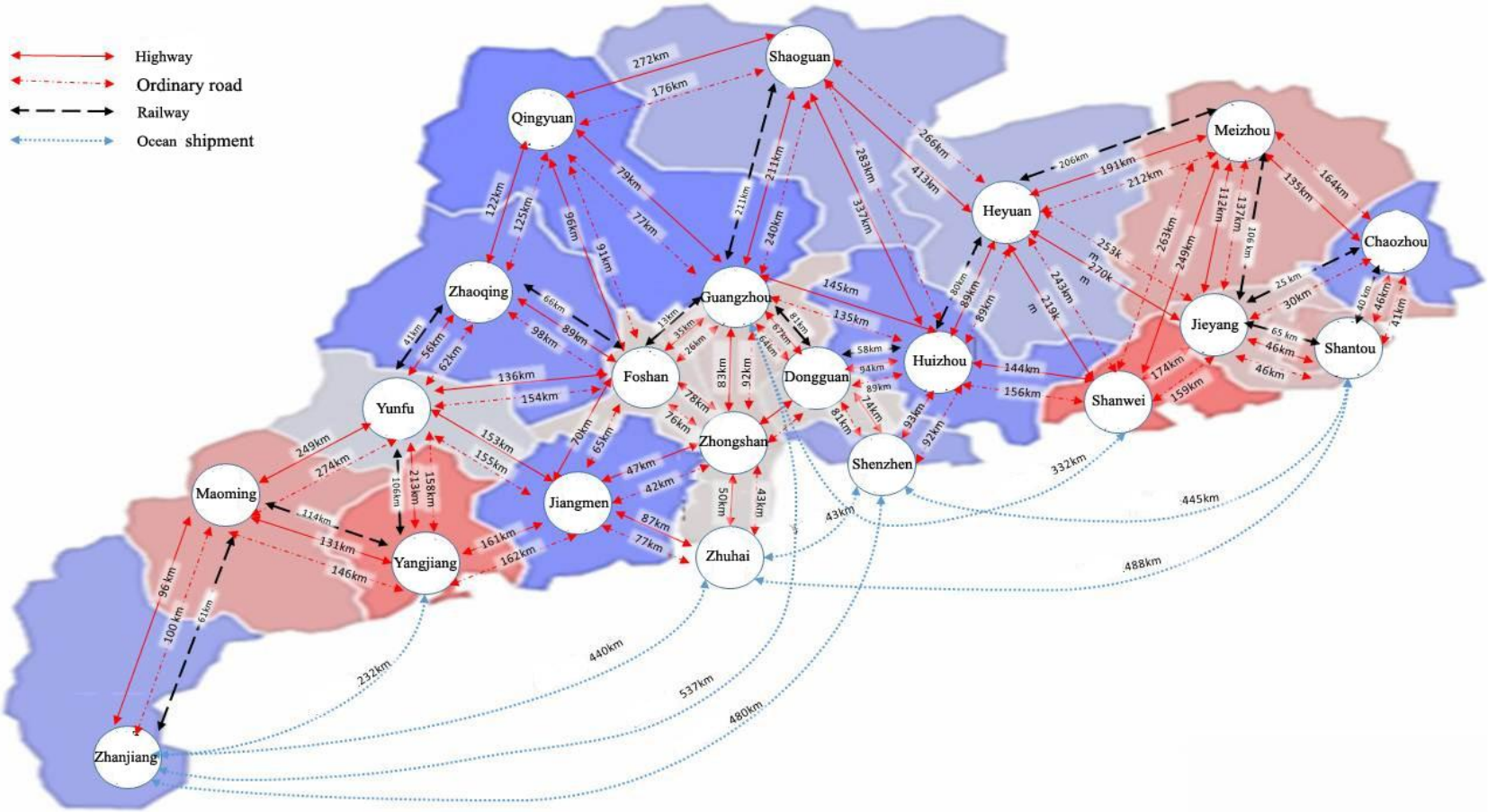


Figure 1. The regional logistics of Pearl River Delta (PRD) in Guangdong province of China (Source: Map Publisher of Guangdong Province. The transportation map of Guangdong Province. 2005.09)

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A. The situation of single objective

When minimizing the total transportation cost (objective function f_1), the optimal solution is shown in Figure 2 where the left side nodes are supply nodes and the right nodes are demand nodes. As we can see, to minimize the total cost, no other transshipment nodes are used, and the supply is first combined together by ocean (from Shanwei to Guangzhou) and railway (from Guangzhou to Dongguan). The optimal solution is shown in Figure 3 when the objective is to minimize the total carbon emission (f_2). From this figure we can see that more transshipment nodes (as shown in the middle column) are added in order to decrease the carbon emission. For example, Zhanjiang is introduced for ocean transportation from Guangzhou to Zhuhai. Heyuan, Meizhou, Jieyang and Shantou are used to deliver goods for Shenzhen by railway and ocean. The results of the two solutions are compared in Table 2, Figure 4 and Figure 5. It is clearly that the ordinary road used in the first solution (minimizing the total transportation cost) is replaced by railway and ocean in the second solution which tries to minimize the total carbon emission. The quantity delivered by environment friendly ways, railway and ocean, is more than five times in solution 2 than in solution 1. While the transportation quantity by highway is similar in both solutions. As a result, comparing with solution 1, the total carbon emission decreases 47.84% and the total transportation cost increases 30.66% in solution 2 because of much longer transportation distance appears when railway and ocean are adopted (as shown in Figure 5).

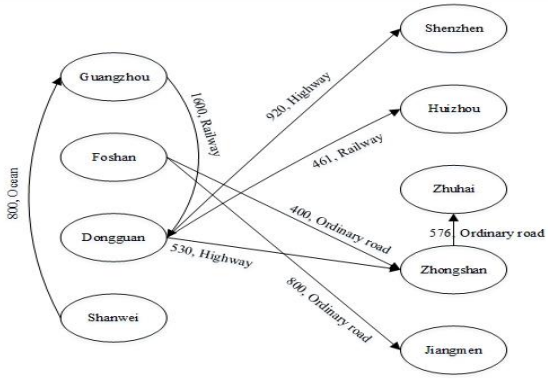


Figure 2. The solution for minimizing total transportation cost

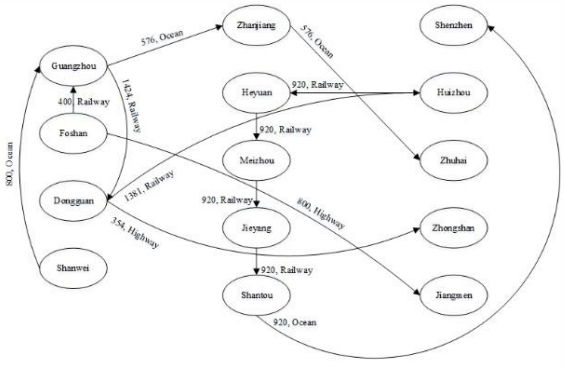


Figure 3. The solution for minimizing total carbon emission

Table 2. THE COMPARISON OF THE TWO SOLUTIONS (COST: YUAN; DISTANCE: KM; CARBON EMISSION: TON; TRANSPORTATION QUANTITY: TON)

	Solutions 1	Solution 2
Minimizing objective	Cost	Emission
Total transportation cost	599,166	782,860
Total carbon emission	174.010	90.765
Transportation distance by ordinary road	184	0
Transportation distance by highway	119	115
Transportation distance by railway	139	609
Transportation distance by ocean	332	1754
Transportation quantity by ordinary road	1776	0
Transportation quantity by highway	1450	1154
Transportation quantity by railway	2061	6885
Transportation quantity by ocean	800	2872

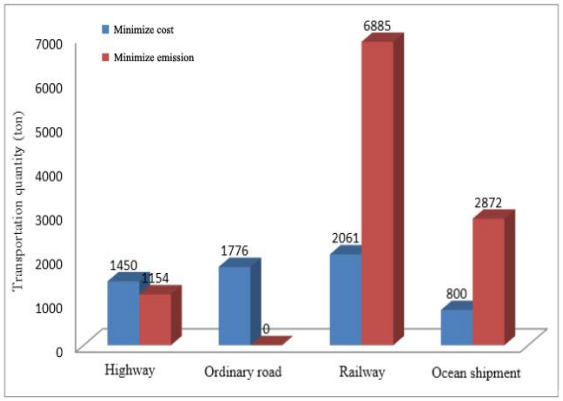


Figure 4. Comparison on transportation quantity

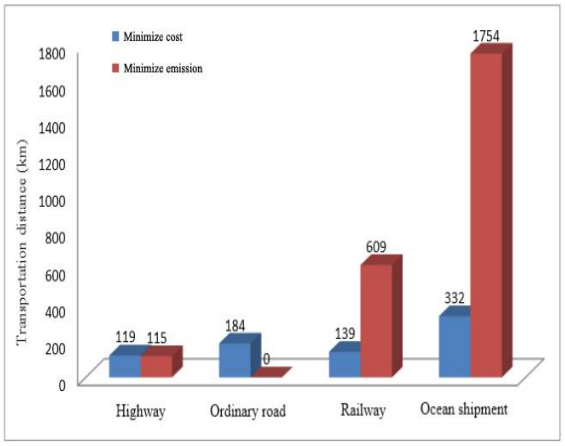


Figure 5. Comparison on transportation distance

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B. The situation of multiple objective

As shown in the analysis of single objective, the total carbon emission may change from 90.765 tons to 174.010 tons due to different transportation solutions. In order to analyze the multiple objective optimization problem, ε-constraint method is used as follows. The total transportation cost is minimized with an additional constraint of total carbon emission ceiling. Five upper bounds of total emission, 95, 115, 135, 155, 175, are added to the constraint conditions respectively. The two-objective pareto noninferior curve is then obtained and shown in Figure 6. To reduce carbon emission, the transportation cost will increase. Furthermore, the decreasing marginal effect phenomenon can be clearly observed in Figure 6, i.e. with the decrease of total carbon emission, to reduce the same amount of emission, the increase of total transportation cost will be higher and higher. It seems that for the case studied, reducing the total carbon emission to around 135 tons is reasonable. Otherwise, the total transportation cost will increase sharply if pursuing a lower carbon emission.

Figure 7 illustrates the transportation volume by different transportation modes under different emission ceilings. With the decrease of emission ceiling, i.e. the carbon emission is controlled tighter and tighter, the transportation quantity by ocean goes up dramatically, while the transportation quantity by ordinary road and highway goes down, especially for ordinary road transportation that produces much more emission. However, the declining trend is not very significant in terms of railway transportation, which means railway can better balance between carbon emission and transportation cost.

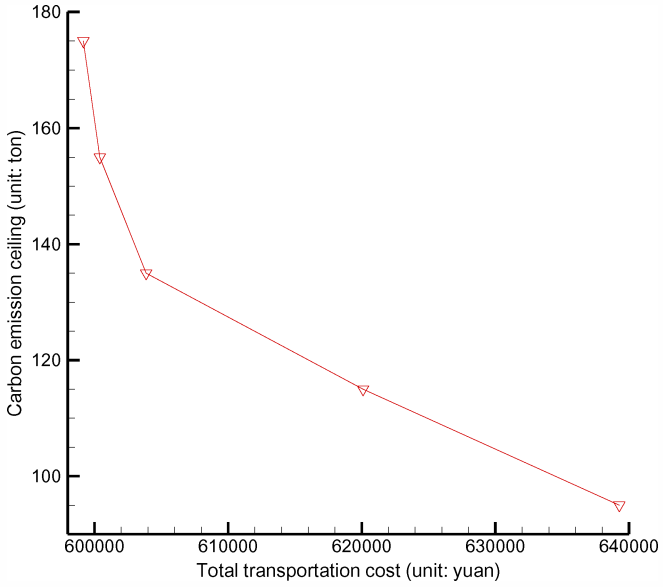


Figure 6. Pareto noninferior curve of multiple objective situation

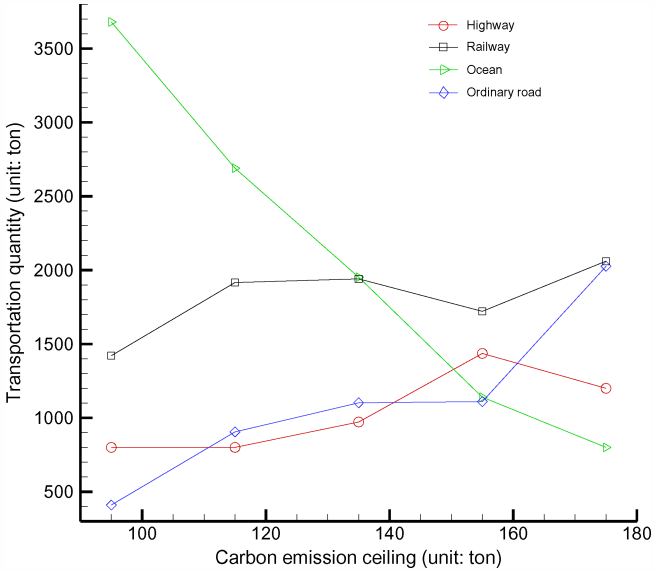


Figure 7. The transportation quantity of different transportation mode under different emission ceiling

IV. CONCLUSION AND DISCUSSION

It becomes more and more important to consider the environmental performance as well as the economic performance when optimizing the multimodal transportation systems. In this paper, a general optimization model is proposed to solve the transportation mode selection and the demand allocation problem. The case study shows that the optimal solution changes a lot when environmental performance (e.g. carbon emission) is regarded as the optimized objective. Railway and ocean shipment are used more frequently instead of ordinary road which is usually a main way for transportation in minimizing the total transportation cost. Comparing minimizing total carbon emission with minimizing total transportation cost respectively, although the transportation cost increases to some extent (about 30%), the carbon emission declines significantly (near 50%). Consequently, it is possible to improve the environmental performance in multimodal transportation under the condition of not greatly increasing the transportation cost. The experimental results of multi-objective situation show that there exist a tradeoff between emission reduction and transportation cost. At the beginning of emission reduction, the increase of transportation cost is not dramatic. However, once the amount of emission reduction exceeds the tradeoff point, it becomes more and more uneconomic to reduce carbon emission due to the sharp increase of transportation cost. With some modification, the model proposed in this paper may also be used to take the transportation time and due dates of demands into consideration, which are the future tasks of this ongoing research.

ACKNOWLEDGEMENT

This research is partially supported by the Natural Science Foundation of China (NSFC contract number 71172057, 71272089, 71472126 and 71572114) and the Young Scholar Supporting Plan of Guangdong Province (contract number

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YQ2014152).

REFERENCES

[1] Bouchery, Y. and Fransoo, J. Cost, “Carbon emissions and modal shift in intermodal network design decisions,” International Journal of Production Economics, 2014, 164(1): 388-399.

[2] Chen, Y.F. and Li, Y.S. “Research on multimodal transport optimization problem based on carbon constraint policies,” Proceedings of the 4th International Conference on Logistics, Informatics and Service Science, 2015: 339-344.

[3] Chen Yingjun, Haishi Ma. Fubo Zhao. “esearch on energy consumption norm of MSA vessels,” China MSA, 2010, 01: 54-56.

[4] Du Lili, Srinivas Peeta, Peng Wei, Dengfeng Sun. “A quantitative and systematic methodology to investigate energy consumption issues in multimodal intercity transportation systems,” International Journal of Transportation Science and Technology, 2015, 4(3): 229-256.

[5] Ehrgott, M. and Ruzika, S. “Improved ε-constraint method for multiobjective programming,” Journal of Optimization Theory and Applications, 2008, 138(3): 375-396.

[6] Guo, S.P. “Internalization of transportation external costs: Impact analysis of logistics company mode and route choices,” Transportation Planning & Technology, 2007, 30(2): 147-165.

[7] Huang Fei-qian and Lin Yu-shan. “Benefit analysis of container sea-railway combined transportation based on low-carbon economy (in Chinese),” Logistics engineering and management, 2011, 33(5): 49-50.

[8] Janic, M. “Assessing some social and environmental effects of transforming an airport into a real multimodal transport node,” Transportation Research Part D: Transport & Environment, 2011, 16(2): 137-149.

[9] Jiang, B., Li, J. and Mao, X. “Container ports multimodal transport in china from the view of low carbon,” Asian Journal of Shipping & Logistics, 2012, 28(3): 321-343.

[10] Kumar P. Phani, Manoranjan Parida, Mansha Swami. “Performance Evaluation of Multimodal Transportation Systems,” Procedia - Social and Behavioral Sciences, 2013, 104(12): 795-804.

[11] Makhloufi R, Cattaruzza D, Meunier F, et al. “Simulation of Mutualized Urban Logistics Systems with Real-time Management,” Transportation Research Procedia, 2015, 6: 365-376.

[12] Song Malin, Shuhong Wang, Ron Fisher. “Transportation, iceberg costs and the adjustment of industrial structure in China,” Transportation Research Part D: Transport and Environment, 2014, 32(10): 278-286.

[13] Wang Zhaohua, Weijun He. “CO2 emissions efficiency and marginal abatement costs of the regional transportation sectors in China,”Transportation Research Part D: Transport and Environment, 2017, 50(11): 83-97.

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